

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 06



Submitted to:

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Submitted by:

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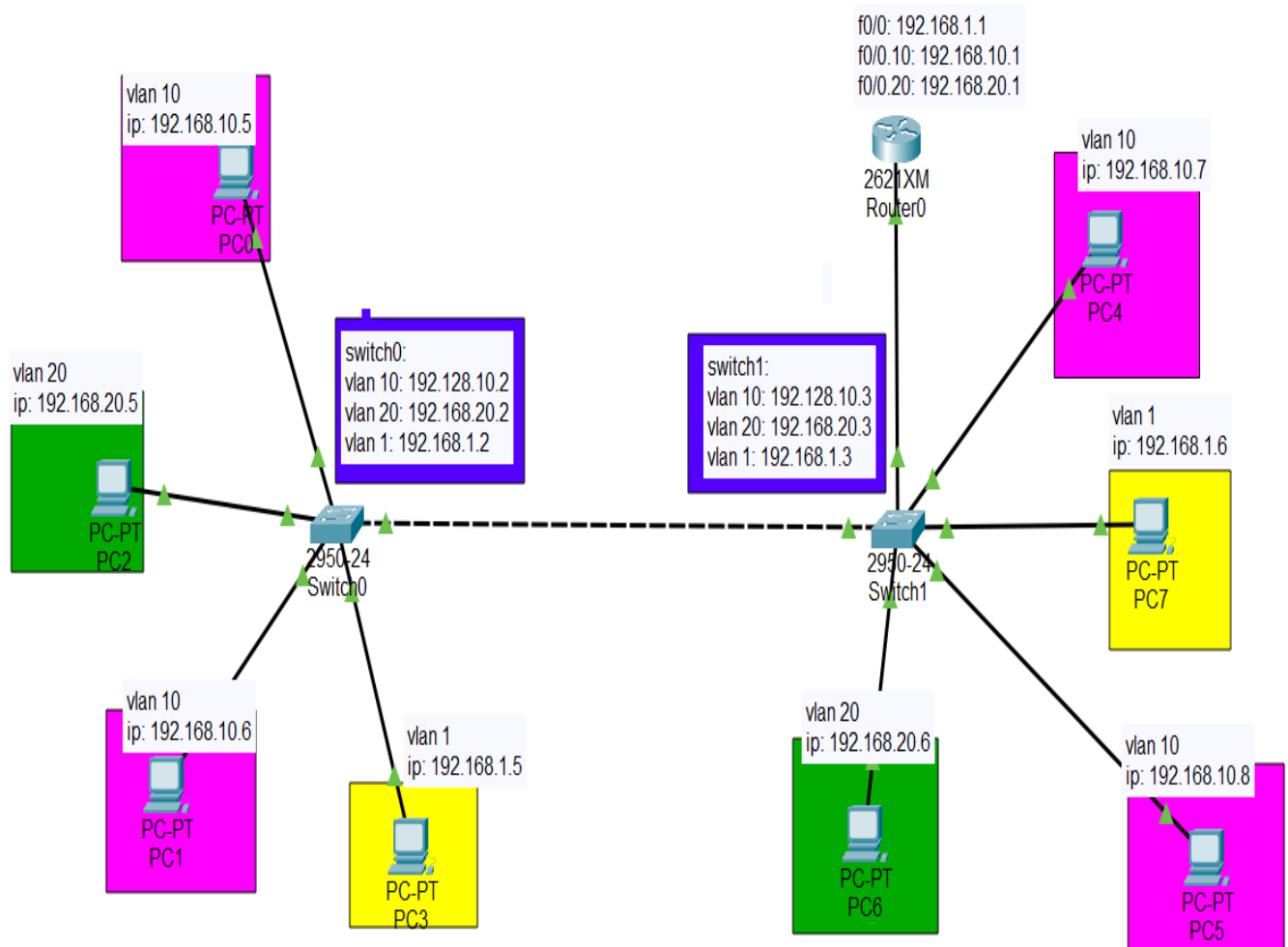
Subject: computer network

Department: BSSE-5-Afternoon

Date: 01-12-2024

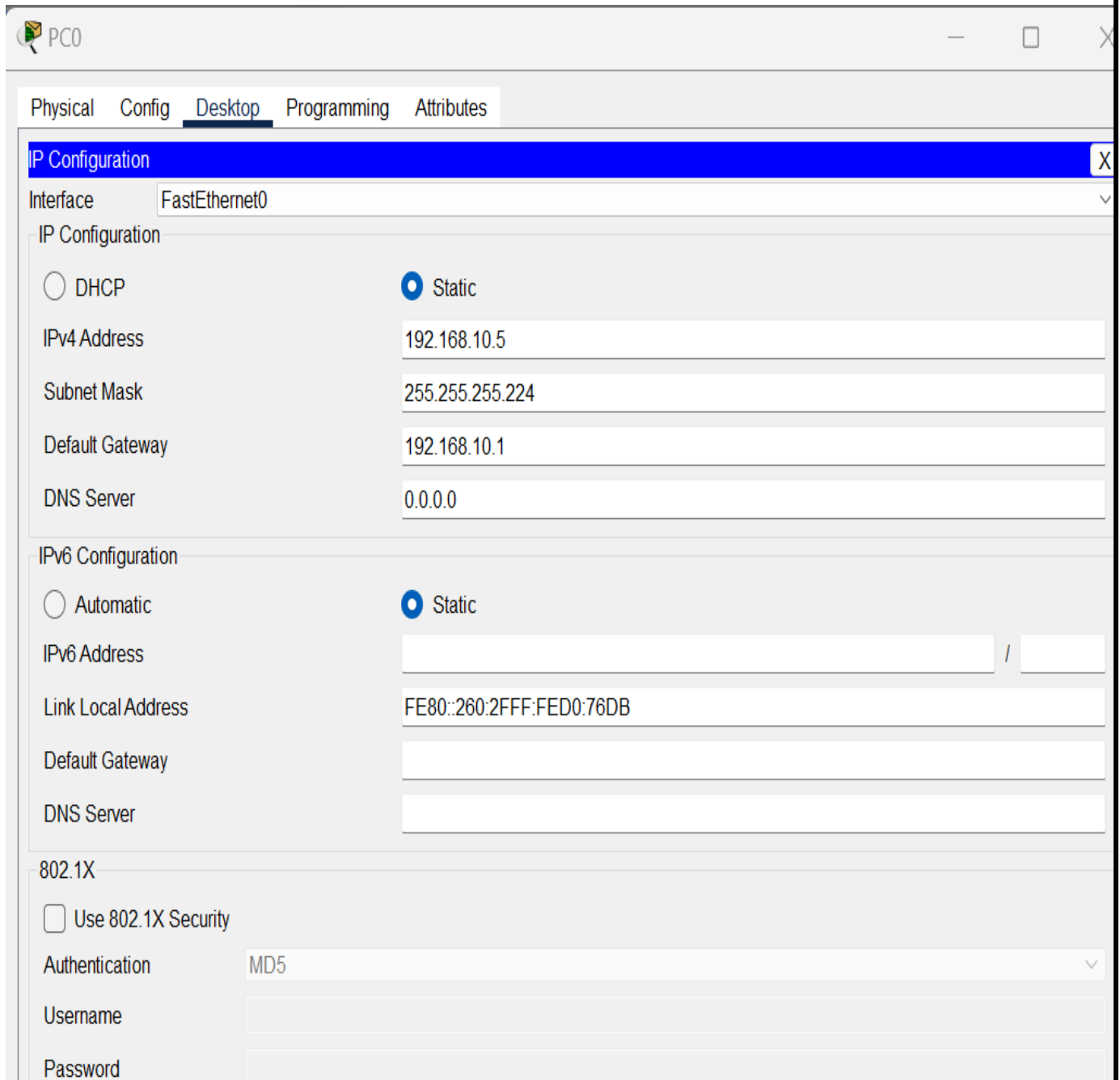
Objectives:

- Create and configure VLANs in Cisco packet tracer.
- Study the Router on a Stick Configuration.
- Learn to Configure native VLAN in Cisco packet tracer and Fix native VLAN mismatch.



Task 1

set IP address and default gateway to pcs.



The screenshot shows the configuration window for PC0 in a network simulator. The 'Desktop' tab is selected, and the 'IP Configuration' section is active. The interface is 'FastEthernet0'. The IP configuration is set to 'Static' with an IPv4 address of 192.168.10.5, a subnet mask of 255.255.255.224, and a default gateway of 192.168.10.1. The DNS server is set to 0.0.0.0. The IPv6 configuration is also set to 'Static', with a link local address of FE80::260:2FFF:FED0:76DB. The 802.1X security section is visible but not configured.

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.5
Subnet Mask	255.255.255.224
Default Gateway	192.168.10.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	/
Link Local Address	FE80::260:2FFF:FED0:76DB
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

Task 2

Creating Vlan

```
ZulqarnainHassan(config)#
ZulqarnainHassan(config)#
ZulqarnainHassan(config)#
ZulqarnainHassan(config)#vlan 10
ZulqarnainHassan(config-vlan)#name voice
ZulqarnainHassan(config-vlan)#exit
ZulqarnainHassan(config)#vlan 20
ZulqarnainHassan(config-vlan)#name sales
ZulqarnainHassan(config-vlan)#exit
ZulqarnainHassan(config)#
```

Allocating multiple ports to a Vlan.

```
ZulqarnainHassan(config)#interface range f0/3-8
ZulqarnainHassan(config-if-range)#sw
ZulqarnainHassan(config-if-range)#switchport m
ZulqarnainHassan(config-if-range)#switchport mode a
ZulqarnainHassan(config-if-range)#switchport mode access
ZulqarnainHassan(config-if-range)#sw
ZulqarnainHassan(config-if-range)#switchport a
ZulqarnainHassan(config-if-range)#switchport access vlan 10
ZulqarnainHassan(config-if-range)#no sh
ZulqarnainHassan(config-if-range)#no shutdown
ZulqarnainHassan(config-if-range)#exit
ZulqarnainHassan(config)#inter
ZulqarnainHassan(config)#interface r
ZulqarnainHassan(config)#interface range f0/9-14
ZulqarnainHassan(config-if-range)#sw
ZulqarnainHassan(config-if-range)#switchport m
ZulqarnainHassan(config-if-range)#switchport mode
ZulqarnainHassan(config-if-range)#switchport mode A
ZulqarnainHassan(config-if-range)#switchport mode Access
ZulqarnainHassan(config-if-range)#SW
ZulqarnainHassan(config-if-range)#SWitchport A
ZulqarnainHassan(config-if-range)#SWitchport Access vlan 20
ZulqarnainHassan(config-if-range)#exit
```

Task3

Trunking port.

```
ZulqarnainHassan(config)#
ZulqarnainHassan(config)#inter
ZulqarnainHassan(config)#interface f0/1
ZulqarnainHassan(config-if)#sw
ZulqarnainHassan(config-if)#switchport m
ZulqarnainHassan(config-if)#switchport mode tr
ZulqarnainHassan(config-if)#switchport mode trunk
ZulqarnainHassan(config-if)#
ZulqarnainHassan(config-if)#exit
ZulqarnainHassan(config)#
```

Task 4

setting up a subinterface on a router.

```
-----
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#interface f0/0
Zulqarnain_SP21379(config-if)#no shutdown

Zulqarnain_SP21379(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Zulqarnain_SP21379(config-if)#
Zulqarnain_SP21379(config-if)#
Zulqarnain_SP21379(config-if)#exit
-----

Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#intr
Zulqarnain_SP21379(config)#inter
Zulqarnain_SP21379(config)#interface f
Zulqarnain_SP21379(config)#interface fastEthernet 0/0.10
Zulqarnain_SP21379(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

Zulqarnain_SP21379(config-subif)#en
Zulqarnain_SP21379(config-subif)#encapsulation dot1q
Zulqarnain_SP21379(config-subif)#encapsulation dot1q 10
Zulqarnain_SP21379(config-subif)#ip ad
Zulqarnain_SP21379(config-subif)#ip address 192.168.10.1 255.255.255.0
Zulqarnain_SP21379(config-subif)#exit
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#inter
Zulqarnain_SP21379(config)#interface f
Zulqarnain_SP21379(config)#interface fastEthernet 0/0.20
Zulqarnain_SP21379(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

Zulqarnain_SP21379(config-subif)#en
Zulqarnain_SP21379(config-subif)#encapsulation d
Zulqarnain_SP21379(config-subif)#encapsulation dot1Q 20
Zulqarnain_SP21379(config-subif)#ip ad
% Incomplete command.
Zulqarnain_SP21379(config-subif)#ip ad
Zulqarnain_SP21379(config-subif)#ip address 192.168.20.1 255.255.255.0
Zulqarnain_SP21379(config-subif)#
Zulqarnain_SP21379(config-subif)#exit
Zulqarnain_SP21379(config)#
-----
```

Assigning IP addresses to vlans.

```
ZulqarnainHassan(config)#
ZulqarnainHassan(config)#
ZulqarnainHassan(config)#interf
ZulqarnainHassan(config)#interface vlan 10
ZulqarnainHassan(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

ZulqarnainHassan(config-if)#ip ad
ZulqarnainHassan(config-if)#ip address 192.168.10.2 255.255.255.224
ZulqarnainHassan(config-if)#exit
ZulqarnainHassan(config)#interface vlan 20
ZulqarnainHassan(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

ZulqarnainHassan(config-if)#ip address 192.168.20.2 255.255.255.224
ZulqarnainHassan(config-if)#exit
ZulqarnainHassan(config)#interface vlan 1
ZulqarnainHassan(config-if)#ip address 192.168.1.2 255.255.255.0
ZulqarnainHassan(config-if)#exit
ZulqarnainHassan(config)#
```